# Investigating the Bias in forecasted HRRR precipitation compared to RTMA, ASOS, and NYS Mesonet

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## INTRODUCTION

- The majority of weather forecasts are created using Numerical Weather Prediction (NWP) models, which assimilate data from multiple sources and compute various physics-based equations to attempt to model the fluid dynamics of the atmosphere and oceans from the current state of the atmosphere to one timestep in the future.
- The conditions from that previously-computed timestep are then inputted to recursively derive the desired forecast. Small inaccuracies of each forecast multiply over time leading to increasing error as the number of timesteps in the future increases. Many of the processes that contribute to precipitation cannot be directly computed using physics equations and must be estimated, therefore making predicting precipitation and precipitation type a challenge even with advancements in NWPs.
- While the problems with NWP precipitation prediction is well known, to quantify the forecasting error and make a case for deep learning post processing, NWP precipitation model output is compared to an analysis dataset.
- To assess the precipitation bias or forecasting error, the High-Resolution-Rapid-Refresh (HRRR) 12 hour forecasted precipitation values are chosen to compared to the Real-Time Mesoscale Analysis (RTMA) dataset.
- This analysis is preformed on the native scale of the RTMA dataset, 2.5 km resolution and on the county level for New York counties.
- Weather station gauge precipitation values are also compared to both RTMA and HRRR values at those points, as gauge precipitation offers the most accurate ground truth in the precipitation that reaches the surface.

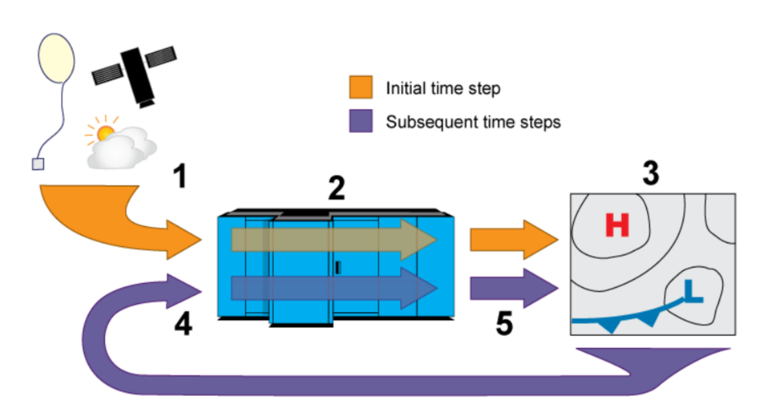


*Figure 1. Simple Representation of NWP processing (NOAA 2023).*

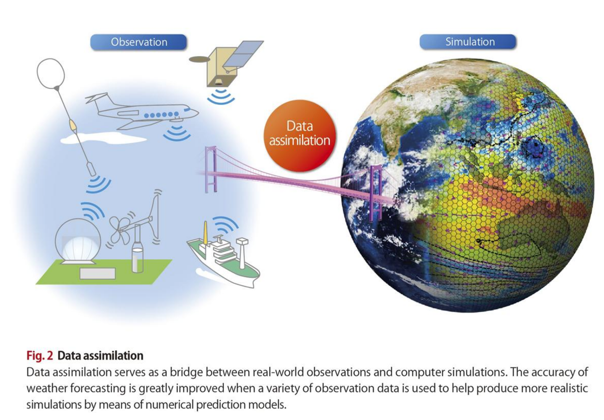

*Figure 2. Data Assimilation model (Miyoshi, 2016)*

## MATERIALS

- NWP Forecast: High-Resolution Rapid-Refresh (HRRR)
  - Resolution: ~ 3000 meter x 3000 meter
  - 12 hour forecasts (forecasts created 12 hours before the forecasted time)
  - Accessed from MesoWest in Zarr format
- Analysis Dataset: Real-Time Mesoscale Analysis (RTMA)
  - Resolution: ~ 2500 meter x 2500 meter
  - Initialized using radar and satellite precipitation estimates
  - Accessed through Google Earth Engine (GEE)
- Observational Datasets:
  - New York State Mesonet (NYSM)
  - Automated Surface Observing System (ASOS)
- Date Range:
  - One year (2021) of data was used due to storage and computational constraints and the mean precipitation per pixel for that year is compared.

## METHODS

- The study area includes New York State and portions of surrounding states.
- HRRR was chosen for its extensive use in analysis literature and easy accessibility of historic model output.
- RTMA was chosen as its one of the few gridded precipitation datasets that has historic data, an extremely high resolution, and accessibility through GEE.
- 2021 was chosen as the analysis year as this is the first year with surface variable data available in Zarr format from MesoWest.
- In ArcGIS Pro, HRRR 2021 mean precipitation was clipped, snapped, and reprojected to match the RTMA 2021 mean precipitation extent and cell size.
- Census State shapefiles from ArcGIS Online were accessed to better visualize the study area. The 2021 mean precipitation for both gridded datasets, NYSM and ASOS points, and the three datasets respective distributions are shown below.

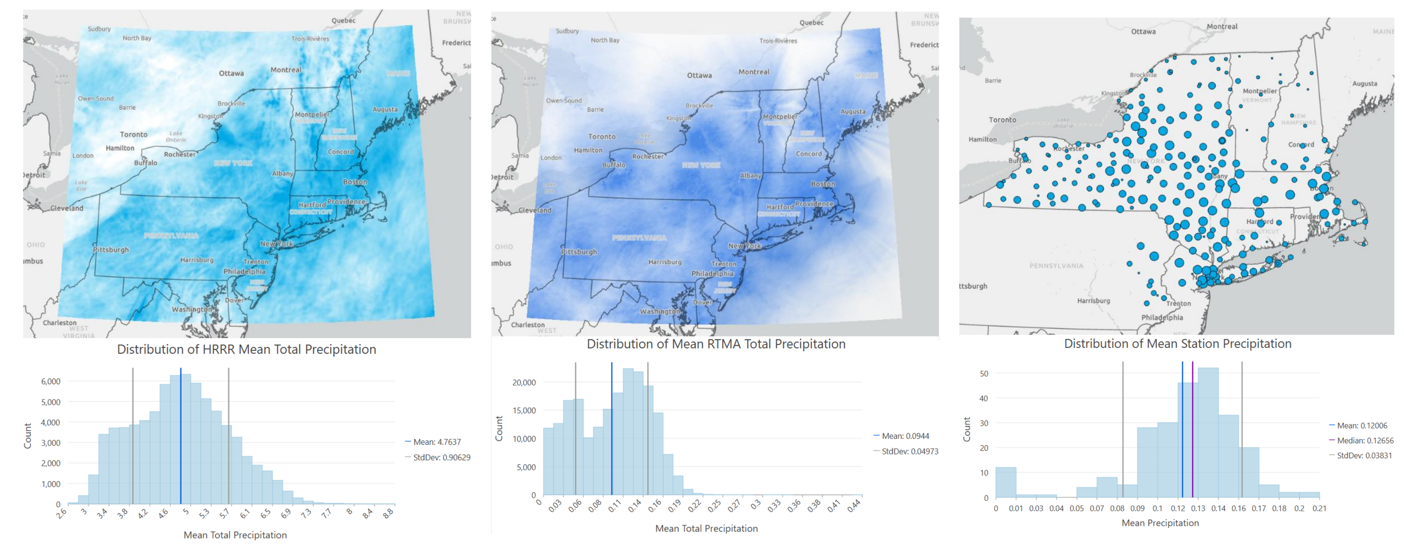


*Figure 3. 2021 Mean HRRR (left) RTMA (middle) and ASOS/NYSM (left) precipitation and respective distributions.*

- To calculate the bias, the raster calculator tool was used to subtract RTMA values

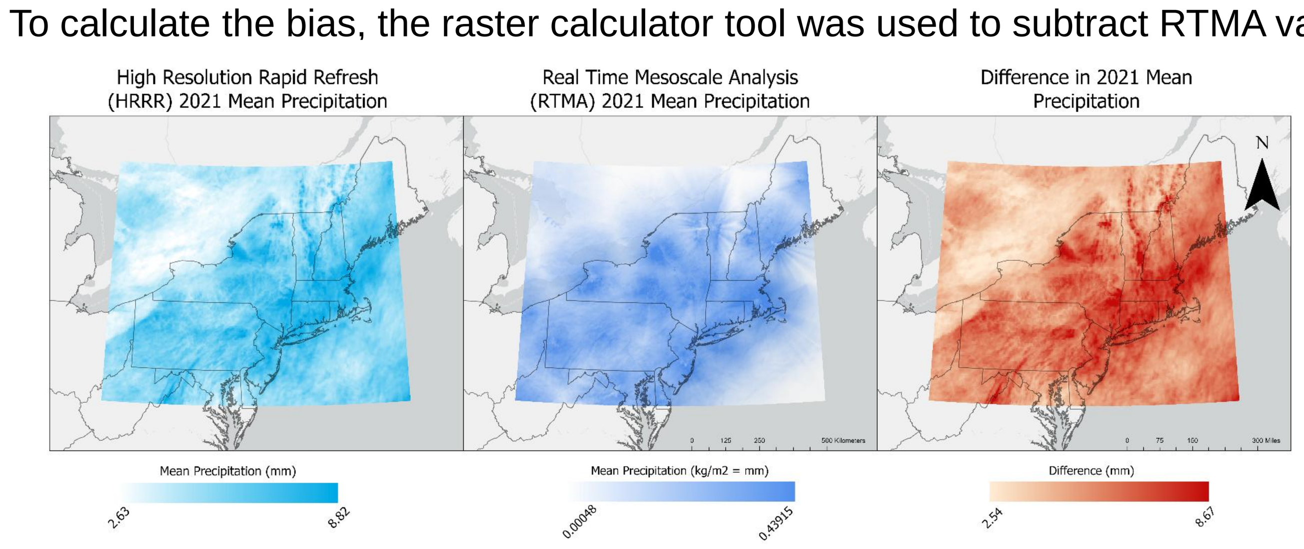


*Figure 4. Bias/Difference between HRRR and RTMA precipitation.*

- Zonal statistics was used to calculate the mean of the mean precipitation in each NY county for clearer visualization of the bias.

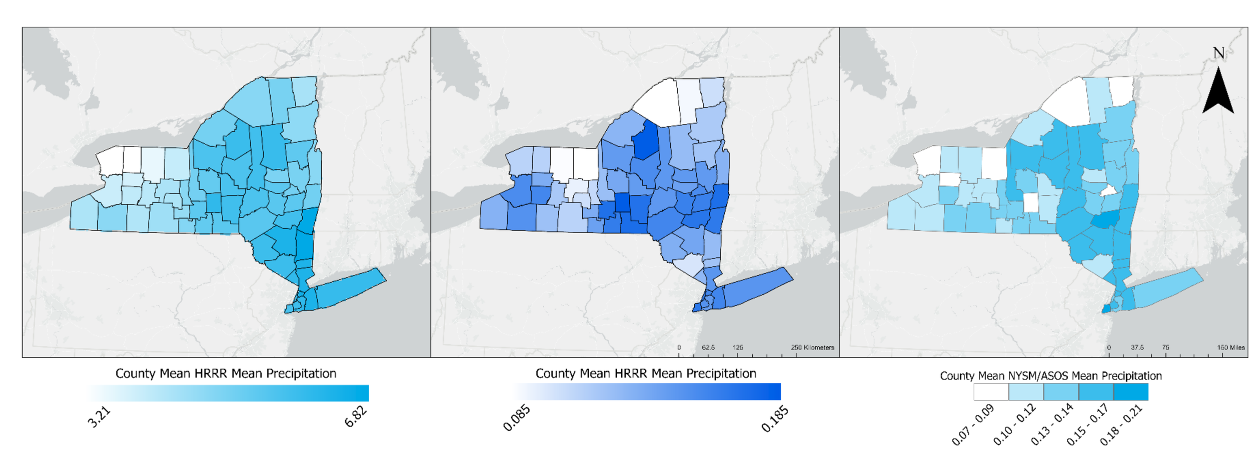


*Figure 5. County level zonal statistics*

- The summary statistics tool was used to calculate the mean precipitation for weather station points in 2021.
- Extract values to points was used to compare weather station point precipitation and gridded dataset values at or near those points.

## RESULTS

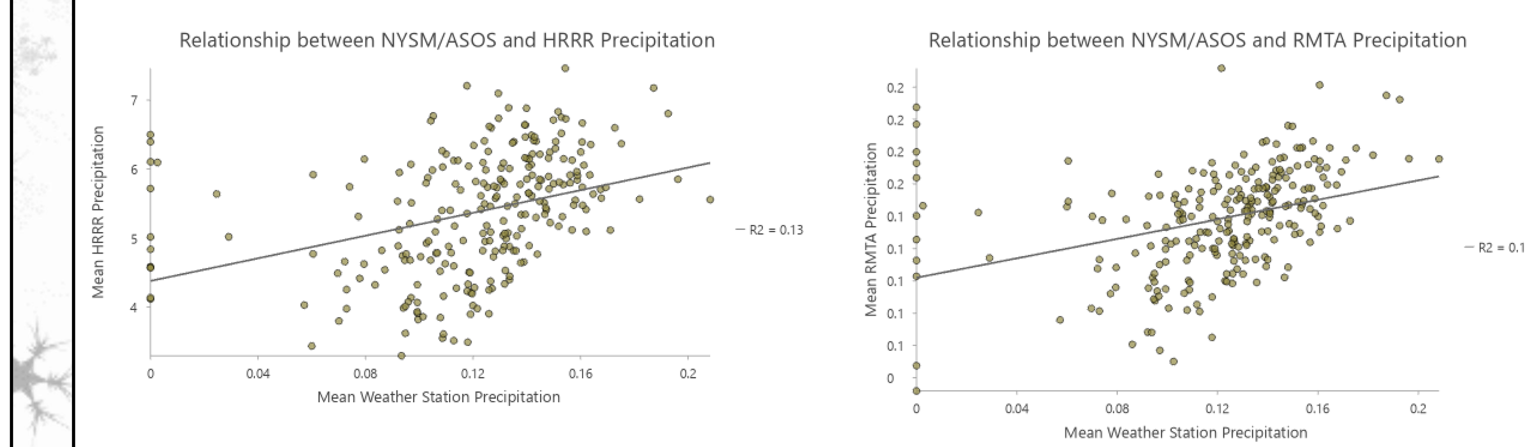


*Figure 6. Relationship between HRRR (left) and RTMA (right) precipitation and weather station precipitation values*

- Figure 6., shows the relationship between HRRR and RTMA precipitation compared to NYSM and ASOS weather station precipitation. Both gridded datasets have a similar level to weather station precipitation, shown in their equal but low $R^2$ values.
  - However, when comparing the distributions in Figure 3., we see that the bimodal RTMA distribution more closely resembles the ASOS/NYSM aggregated distribution.
  - Ranges in Figures 3. and 6., show that the RTMA dataset is more similar, as well when compared to the weather station points.
- The difference/bias map in Figure 3., shows a difference range of 2.54 to 8.67 mm.
  - These positive values indicate the tendency for HRRR to overpredict precipitation for 2021 forecasts.
- On the county level, the same comparison is evident, with counties showing more similar values between each other in the HRRR dataset; and low and high clusters showing similarity between the RTMA and weather station county mean values.
- Radially shaped artifacts are visible in the RTMA dataset, most likely the result of data collection through sweeping radar methods.
- HRRR data seems to be heavily influenced by orography, or the study of topographic reliefs from mountains and valleys as evident in Figure 3.

## FUTURE WORK

Future work will involve:

- Investigating to see if the artifacts in the RTMA dataset reduce the validity of the data as a ground truth for post processing.
- Comparison of a greater number of years between datasets to determine long term bias.
- Comparison of multiple forecast lead times, beyond 12 hours, to determine if bias trends change with lead time and the most crucial forecast time to post process.
- Possible interpolation of weather station points or optimal interpolation of weather station points and RTMA data to account for the bias between the two datasets and creation of an accurate gridded observational dataset.
- Use of datasets for post-processing of HRRR forecasted precipitation to

## References


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